

E-Commerce Data Warehouse

Schema Reference · Star Schema Design · Joe Laniado · June 2026

PK Primary Key	FK Foreign Key	PK/FK Both	★ Key project columns	Fact tables	Dimension tables
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SCHEMA OVERVIEW

FACT_ORDERS	Fact	order_line_id	customer_id, product_id, seller_id, location_id, campaign_id, payment_id, fulfillment_id, delivery_person_id, date_id	discount_percentage ★, customer_rating ★, order_status, return_flag, sentiment_score	One row per order line item
FACT_RETURNS	Fact	order_line_id	customer_id, product_id, location_id, payment_id, fulfillment_id, date_id	refund_amount, days_to_return, customer_rating	One row per returned item
FACT_MARKETING_SPEND	Fact	marketing_id	campaign_id, channel_id, date_id	spend_amount, conversions, revenue_generated	One row per campaign-channel-date
FACT_FULFILLMENT_PERFORMANCE	Fact (aggregated)	fulfillment_id	location_id, delivery_person_id, date_id	sla_breach_rate, return_rate, on_time_deliveries	Pre-aggregated KPIs
FACT_CUSTOMER_RFM	Fact (aggregated)	customer_id	customer_id → DIM_CUSTOMER	customer_segment, rfm_score, monetary_value	One row per customer, pre-computed
DIM_CUSTOMER	Dimension	customer_id	location_id	income_bracket, gender, registration_date	One row per customer
DIM_PRODUCT	Dimension	product_id	—	category, sub_category, list_price	One row per product
DIM_SELLER	Dimension	seller_id	location_id	rating, category_focus, ifsc_code	One row per seller
DIM_LOCATION	Dimension (shared)	location_id	—	state, city, region, area_type	One row per location. Shared across 5 tables.
DIM_CAMPAIGN	Dimension	campaign_id	channel_id	campaign_type, budget, objective	One row per campaign
DIM_CHANNEL	Dimension	channel_id	—	channel_name, channel_type	One row per channel

DIM_PAYMENT	Dimension	<i>payment_id</i>	—	<i>payment_method, payment_provider</i>	<i>One row per payment method</i>
DIM_FULFILLMENT	Dimension	<i>fulfillment_id</i>	—	<i>shipping_method, service_level, delivery_sla_days</i>	<i>One row per fulfillment type</i>
DIM_DELIVERY_PERSON	Dimension	<i>delivery_person_id</i>	<i>location_id</i>	<i>employment_type, vehicle_type</i>	<i>One row per delivery person</i>
DIM_CALENDAR	Dimension	<i>date_id</i>	—	<i>month_name, quarter, year, is_weekend</i>	<i>One row per date</i>

FACT TABLES — TRANSACTIONAL & AGGREGATED EVENT DATA

FACT_ORDERS (cols 1-18)		
order_line_id	INT	PK
order_id	INT	
date_id	INT	FK
order_date	DATE	
customer_id	INT	FK
product_id	INT	FK
category	VARCHAR	
seller_id	INT	FK
seller_rating	DECIMAL	
location_id	INT	FK
delivery_person_id	INT	FK
payment_id	INT	FK
campaign_id	INT	FK
fulfillment_id	INT	FK
quantity	INT	
unit_price	DECIMAL	
discount_percentage	DECIMAL	
tax_percentage	DECIMAL	

FACT_ORDERS (cols 19-35)		
shipping_fee	DECIMAL	
expected_delivery_date	DATE	
actual_delivery_date	DATE	
delivery_rating	DECIMAL	
order_status	VARCHAR	
return_flag	BOOLEAN	
refund_amount	DECIMAL	
return_date	DATE	
customer_rating	DECIMAL	
review_text	TEXT	
sentiment_score	DECIMAL	
delivery_delay_days	INT	
delivery_sla_days	INT	
gross_amount	DECIMAL	
discount_amount	DECIMAL	
tax_amount	DECIMAL	
net_amount	DECIMAL	

FACT_RETURNS		
order_line_id	INT	PK/FK
order_id	INT	
date_id	INT	FK
customer_id	INT	FK
product_id	INT	FK
location_id	INT	FK
payment_id	INT	FK
fulfillment_id	INT	FK
quantity	INT	
unit_price	DECIMAL	
refund_amount	DECIMAL	
review_text	TEXT	
days_to_return	INT	
restocking_fee	DECIMAL	
customer_rating	DECIMAL	
sentiment_score	DECIMAL	

FACT_CUSTOMER_RFM		
customer_id	INT	PK/FK
recency_days	INT	
frequency_orders	INT	
monetary_value	DECIMAL	
avg_order_value	DECIMAL	
first_purchase_date	DATE	
last_purchase_date	DATE	
customer_lifetime_days	INT	
rfm_score	DECIMAL	
customer_segment	VARCHAR	

FACT_MARKETING_SPEND		
marketing_id	INT	PK
date_id	INT	FK
campaign_id	INT	FK
channel_id	INT	FK
spend_amount	DECIMAL	
impressions	INT	
clicks	INT	
conversions	INT	
revenue_generated	DECIMAL	

FACT_FULFILLMENT_PERFORMANC E		
date_id	INT	FK
fulfillment_id	INT	PK/FK
location_id	INT	FK
delivery_person_id	INT	FK
total_deliveries	INT	
completed_deliveries	INT	
returned_deliveries	INT	
avg_delivery_delay	DECIMAL	
avg_delivery_rating	DECIMAL	
total_shipping_cost	DECIMAL	
on_time_deliveries	INT	
late_deliveries	INT	
sla_breach_rate	DECIMAL	
return_rate	DECIMAL	

★ Highlighted columns indicate the primary variables for this project: discount_percentage (group definition) and customer_rating (outcome metric). Both columns reside in FACT_ORDERS — no joins required for the core A/B test.

DIMENSION TABLES — DESCRIPTIVE CONTEXT

DIM_CUSTOMER		
customer_id	INT	PK
first_name	VARCHAR	
last_name	VARCHAR	
email	VARCHAR	
phone_number	VARCHAR	
gender	VARCHAR	
date_of_birth	DATE	
registration_date	DATE	
income_bracket	VARCHAR	
marital_status	VARCHAR	
location_id	INT	FK
upi_id	VARCHAR	
credit_card_number	VARCHAR	

DIM_PRODUCT		
product_id	INT	PK
product_name	VARCHAR	
brand	VARCHAR	
category	VARCHAR	
sub_category	VARCHAR	
list_price	DECIMAL	

DIM_SELLER		
seller_id	INT	PK
seller_name	VARCHAR	
email	VARCHAR	
phone_number	VARCHAR	
join_date	DATE	
location_id	INT	FK
rating	DECIMAL	
category_focus	VARCHAR	
bank_name	VARCHAR	
bank_account_number	VARCHAR	
ifsc_code	VARCHAR	

DIM_LOCATION		
location_id	INT	PK
state	VARCHAR	
city	VARCHAR	
postal_code	VARCHAR	
region	VARCHAR	
latitude	DECIMAL	
longitude	DECIMAL	
area_type	VARCHAR	
location_category	VARCHAR	

DIM_CAMPAIGN		
campaign_id	INT	PK
campaign_name	VARCHAR	
campaign_type	VARCHAR	
channel_id	INT	FK
start_date	DATE	
end_date	DATE	
budget	DECIMAL	
objective	VARCHAR	

DIM_CHANNEL		
channel_id	INT	PK
channel_name	VARCHAR	
channel_type	VARCHAR	
description	TEXT	

DIM_PAYMENT		
payment_id	INT	PK
payment_method	VARCHAR	
payment_provider	VARCHAR	
description	TEXT	

DIM_FULFILLMENT		
fulfillment_id	INT	PK
shipping_method	VARCHAR	
service_level	VARCHAR	
delivery_slas	INT	
base_shipping_cost	DECIMAL	

DIM_DELIVERY_PERSON		
delivery_person_id	INT	PK
delivery_person_name	VARCHAR	
phone_number	VARCHAR	
gender	VARCHAR	
date_of_joining	DATE	
employment_type	VARCHAR	
vehicle_type	VARCHAR	
location_id	INT	FK

DIM_CALENDAR		
date_id	INT	PK
full_date	DATE	
day	INT	
month	INT	
month_name	VARCHAR	
quarter	INT	
year	INT	
day_name	VARCHAR	
is_weekend	BOOLEAN	

DIM_LOCATION is a shared dimension — referenced by FACT_ORDERS, FACT_FULFILLMENT_PERFORMANCE, DIM_CUSTOMER, DIM_SELLER, and DIM_DELIVERY_PERSON. One location table serves the entire schema.

KEY RELATIONSHIPS

FACT_ORDERS	customer_id	DIM_CUSTOMER	Many orders → one customer
FACT_ORDERS	product_id	DIM_PRODUCT	Many orders → one product
FACT_ORDERS	seller_id	DIM_SELLER	Many orders → one seller
FACT_ORDERS	location_id	DIM_LOCATION	Many orders → one location
FACT_ORDERS	campaign_id	DIM_CAMPAIGN	Many orders → one campaign (nullable)
FACT_ORDERS	payment_id	DIM_PAYMENT	Many orders → one payment method
FACT_ORDERS	fulfillment_id	DIM_FULFILLMENT	Many orders → one fulfillment type
FACT_ORDERS	delivery_person_id	DIM_DELIVERY_PERSON	Many orders → one delivery person
FACT_ORDERS	date_id	DIM_CALENDAR	Many orders → one date
FACT_RETURNS	order_line_id	FACT_ORDERS	One return → one order line (subset)

<i>FACT_CUSTOMER_RFM</i>	<i>customer_id</i>	<i>DIM_CUSTOMER</i>	<i>One RFM record → one customer</i>
<i>FACT_MARKETING_SPEND</i>	<i>campaign_id</i>	<i>DIM_CAMPAIGN</i>	<i>Many spend records → one campaign</i>
<i>DIM_CAMPAIGN</i>	<i>channel_id</i>	<i>DIM_CHANNEL</i>	<i>Many campaigns → one channel</i>
<i>DIM_CUSTOMER</i>	<i>location_id</i>	<i>DIM_LOCATION</i>	<i>Many customers → one location</i>
<i>DIM_SELLER</i>	<i>location_id</i>	<i>DIM_LOCATION</i>	<i>Many sellers → one location</i>

Data Warehouse Schema Reference · E-Commerce Data Warehouse Dataset (Kaggle) · Synthetic dataset · Joe Laniado · June 2026